E-Commerce Order Management System

Business Requirements Document (BRD)

Version: 1.0 | **Prepared By:** Business Analyst | **Date:** March 15, 2026
Status: Requirements Gathering Phase

Why We're Building This

Right now, our online store works — but barely. Every order requires someone to manually verify and enter it into the system, customers have no idea where their package is after they've paid, and our single payment gateway fails about 12% of the time. That's a real problem.

This document lays out what we want to build: a modern, automated e-commerce order management system that handles everything from the moment a customer adds something to their cart, all the way through delivery confirmation. Our goal is to cut order processing time by 60%, make customers happier, and set ourselves up to grow — sustainably.

We're targeting a platform that can handle 10,000+ transactions a day, with real-time inventory, multiple payment options, and smart automation throughout. If we do this right, we're looking at a 180% ROI within 18 months.

What We're Trying to Achieve

Our Primary Goals

1. **Boost our conversion rate** from 45% to 70% within 6 months — by making checkout smoother and faster
2. **Speed up order processing** from 4 hours down to 30 minutes — through automation

3. **Improve customer satisfaction** — targeting an NPS of 60+ through better tracking and communication
4. **Handle 3x more orders** without needing to triple our staff
5. **Cut fulfillment errors** from 8% down to under 2% — so customers get exactly what they ordered

Secondary Goals

- Use analytics to make smarter business decisions
- Improve how quickly we move inventory by 25%
- Reduce cart abandonment from 70% to 50%
- Build a foundation that can grow into omnichannel retail in the future

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Where We Are Today (And Why It Hurts)

Here's an honest look at our current situation:

- **Orders are processed manually** — staff have to verify each one by hand and enter it into the fulfillment system. This is slow and error-prone.
- **Inventory is always a guessing game** — there's no live stock tracking, so we regularly oversell products we don't actually have.
- **Our payment setup is fragile** — one payment gateway with a 12% failure rate isn't good enough.
- **Customers are left in the dark** — once an order is placed, customers get no updates until it shows up at their door (or doesn't).
- **Our systems don't talk to each other** — the website, inventory system, and shipping tools are all separate, which creates data gaps everywhere.

The Stakes

If we don't fix this, the consequences are real:

- We're losing ground to competitors who already have better systems

- About 15% of our customers leave each year because of poor experience
- We can't keep up during peak seasons like Diwali or major sale events
- We're losing roughly $500,000 per year just from cart abandonment

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What We're Building

Customer-Facing Features

Everything a shopper needs for a great experience:

- Easy sign-up via email or social login (Google, Facebook)
- A fully searchable product catalog with filters, sorting, and clear availability info
- Rich product pages with zoom-able images, variants, reviews, and estimated delivery dates
- A smooth shopping cart that updates in real-time without page reloads
- Wishlists for saving products for later
- Guest checkout — no forced registration
- Multiple saved delivery addresses
- Multiple payment options: UPI, Credit/Debit Card, Net Banking, Wallets
- Live order tracking with SMS and email updates at every stage
- Full order history with easy reordering

Admin and Operations Features

Tools that make our team's lives easier:

- A central order management dashboard showing everything at a glance
- Inventory management with low-stock alerts and real-time updates
- Automated shipping label generation with courier partner integrations (Delhivery, BlueDart)
- Bulk order status updates
- Promotional campaign management

- Returns and refund processing
- Role-based access so each team member sees only what they need
- A powerful analytics dashboard for sales, traffic, and customer behavior

What's NOT in This Version

Some things we've consciously left for later phases:

- Native iOS/Android apps → Phase 2
- Multi-language support → Phase 2
- Integration with offline stores → Phase 3
- Marketplace for third-party sellers → Future consideration
- Cryptocurrency payments, AR/VR product visualization, voice ordering → Not planned currently

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What We're Assuming (And What Limits Us)

Assumptions We've Made

1. Our existing product catalog can be migrated to the new system
2. Payment providers (Razorpay, PayU, Paytm) will give us sandbox environments to test with
3. Shipping partners (Delhivery, BlueDart) have APIs we can connect to
4. SSL certificate and domain name are already sorted
5. Our IT team will handle the hosting setup on AWS or Azure
6. Customer data from the legacy system can be migrated cleanly

Real Constraints We're Working Within

- **Budget:** ₹50 lakhs (~$60,000 USD) — that's a firm ceiling
- **Timeline:** We need to go live within 6 months to catch the festive season
- **Tech stack:** We're staying with Node.js (backend) and React (frontend) for consistency
- **Team size:** 5 developers + 2 QA engineers — no more

- **Compliance:** Must follow RBI payment guidelines and Indian data protection laws

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Who This Affects

Person	Role	What They Need From This
CEO	Project Sponsor	Final approval, budget sign-off
VP Operations	Business Owner	Requirements clarity, UAT sign-off
Marketing Head	Key Stakeholder	Promotional tools, analytics
Customer Support Lead	End User	Smooth order tools, easy support workflows
IT Manager	Technical Owner	Architecture decisions, infrastructure
Warehouse Manager	End User	Simple fulfillment workflow
Finance Controller	Key Stakeholder	Payment reconciliation, clean reports
End Customers	Primary User	A seamless shopping and delivery experience

Table 1: Stakeholder identification and roles

What the System Must Do — Functional Requirements

FR-1: Signing Up and Logging In

- Customers can register with an email/password or sign in via Google or Facebook
- Email verification happens automatically on sign-up
- Passwords must be strong (8+ characters, mix of letters, numbers, and special characters)

- "Forgot Password" works via OTP
- After 5 failed login attempts, the account locks for 30 minutes or until reset by email
- Session timeout after 30 minutes of inactivity

Priority: Must Have

User Story: As a new customer, I want to register an account so that I can save my preferences and track my orders.

Acceptance Criteria:

- User can successfully register with valid email and password
- Duplicate email registrations are prevented with appropriate error message
- Password reset link expires after 24 hours
- Social login redirects correctly and creates user profile

FR-2: Finding Products

- Products display in a clean grid or list with images, prices, and ratings
- Search has autocomplete to help customers find things quickly
- Filters work by category, price, brand, rating, and availability — and they can be combined
- Products clearly show "Out of Stock" when unavailable
- Badges like "New," "Sale," and "Bestseller" help customers navigate
- Pagination with configurable items per page

Priority: Must Have

User Story: As a customer, I want to easily find products I'm looking for so that I can make quick purchase decisions.

Acceptance Criteria:

- Search returns relevant results within 2 seconds
- Filters can be applied in combination
- Product count updates dynamically when filters applied
- Sort options work correctly across all pages

FR-3: Product Detail Pages

- High-quality images with zoom, multiple angles
- Full descriptions, specs, and real-time stock availability
- Price displays with strikethrough on discounted items
- Product variants (size, color) each have their own SKU
- Estimated delivery date shows based on the customer's pincode
- Related product suggestions at the bottom
- Aggregate ratings and individual customer reviews
- "Add to Cart" and "Buy Now" buttons

Priority: Must Have

User Story: As a customer, I want detailed product information so that I can make informed purchase decisions.

Acceptance Criteria:

- Product images load within 1 second on 4G connection
- Stock status updates in real-time
- Variant selection updates price and availability correctly
- Reviews are sorted by most helpful/recent

FR-4: Shopping Cart

- Add, update, and remove items — all without page reload
- Cart persists for logged-in users across sessions
- Guest carts automatically merge when the user logs in
- Promo code field with real-time validation
- Cart shows full price breakdown including taxes and shipping
- Item count badge displayed in header
- Stock availability validated before checkout

Priority: Must Have

User Story: As a customer, I want to add items to cart and modify quantities so that I can purchase multiple products together.

Acceptance Criteria:

- Cart updates are reflected immediately without page reload
- Out-of-stock items show appropriate message

- Promo code validation happens in real-time
- Cart total calculations are accurate including taxes

FR-5: Checkout Process

- Completes in a maximum of 3 steps
- Guest checkout option without mandatory registration
- Address validation confirms delivery serviceability instantly
- Full GST breakdown displayed before payment
- Multiple payment options (UPI, Credit/Debit Card, Net Banking, Wallets)
- Shipping charges calculated based on delivery location
- Payment gateway redirect happens within 2 seconds
- Handles payment success, failure, and pending states

Priority: Must Have

User Story: As a customer, I want a smooth checkout experience so that I can complete my purchase quickly.

Acceptance Criteria:

- Checkout process completes in maximum 3 steps
- Address validation provides real-time feedback
- Payment gateway redirection happens within 2 seconds
- Order confirmation email sent within 1 minute of successful payment

FR-6: Order Tracking for Customers

- Order confirmation page and email with invoice appear immediately after payment
- SMS/email updates sent at every status change: Confirmed → Shipped → Out for Delivery → Delivered
- Customers can cancel within 24 hours if not yet shipped
- Returns/exchanges can be initiated directly from order history
- One-click reorder from any past order
- Order history with date range and status filters
- Real-time tracking with courier partner link

Priority: Must Have

User Story: As a customer, I want to track my orders and view order history so that I know when my products will arrive.

Acceptance Criteria:

- Order status reflects actual fulfillment status
- Notifications sent within 5 minutes of status change
- Order cancellation processes refund automatically
- Tracking information links to courier partner website

FR-7: Order Management for Operations

- Dashboard shows all orders filtered by status (New, Processing, Shipped, Delivered, Cancelled)
- Every status change is time-stamped for audit purposes
- Shipping labels are generated automatically via API when an order ships
- Express or VIP orders are flagged for priority handling
- Bulk status updates can process 100+ orders in under 30 seconds
- Order search by ID, customer name, phone, email
- Picking lists generated for warehouse staff
- Integration with shipping partners for automated workflows

Priority: Must Have

User Story: As an operations manager, I want to efficiently manage all orders so that I can ensure timely fulfillment.

Acceptance Criteria:

- Dashboard loads all data within 3 seconds
- Status updates trigger customer notifications automatically
- Bulk updates process 100+ orders within 30 seconds
- Integration with shipping API generates AWB number instantly

FR-8: Inventory Management

- Stock levels update in real-time the moment an order is placed or cancelled

- Low-stock alerts fire via email within 5 minutes of hitting the threshold
- The system reserves inventory during checkout — and releases it after 15 minutes if abandoned
- Multi-warehouse tracking is supported
- A full audit trail is maintained for every inventory movement
- Inventory reports available (current stock, stock movement, aging)
- Manual inventory adjustments with reason codes

Priority: Must Have

User Story: As an inventory manager, I want real-time stock visibility so that I can prevent overselling and stockouts.

Acceptance Criteria:

- Inventory updates are atomic (no race conditions)
- Stock alerts sent via email within 5 minutes
- Inventory reports generate within 10 seconds
- Reserved inventory releases after 15 minutes if checkout abandoned

FR-9: Payment Processing

- Integrated with at least 3 gateways: Razorpay, PayU, Paytm
- Supports UPI, Credit/Debit Cards, Net Banking, and Mobile Wallets
- Fully PCI-DSS compliant — card details are never stored
- Refunds are automatically processed to the original payment method
- Payment receipts include a full GST breakdown
- Payment transaction log maintained for reconciliation
- Real-time payment status display

Priority: Must Have

User Story: As a customer, I want secure and convenient payment options so that I can complete my purchase with confidence.

Acceptance Criteria:

- Payment gateway load balancing ensures 99.5% uptime
- Payment confirmation received within 30 seconds
- Refunds processed to original payment method within 24 hours
- Transaction failures provide user-friendly error messages

FR-10: Reviews and Ratings

- Only verified buyers can leave reviews (within 90 days of delivery)
- 5-star rating system with text reviews (minimum 20 characters) and optional images
- Reviews go through admin approval before going live
- Review request email sent automatically 7 days after delivery
- "Helpful" voting lets useful reviews surface to the top
- Reviews can be filtered by rating
- Aggregate rating and total review count displayed

Priority: Should Have

User Story: As a customer, I want to read and write product reviews so that I can make better purchase decisions.

Acceptance Criteria:

- Only verified purchasers can submit reviews
- Reviews display within 24 hours of admin approval
- Aggregate rating updates immediately after review approval
- Helpful votes counted accurately

FR-11: Analytics Dashboard

- Real-time display of daily sales, orders, average order value, and conversion rate
- Sales trend charts for last 7, 30, and 90 days
- Category-wise sales, top products, traffic sources, and cart abandonment funnel
- Reports can be exported to Excel or PDF
- Dashboard loads in under 3 seconds and refreshes every 5 minutes
- Customer acquisition metrics
- Date range selection for custom reports

Priority: Should Have

User Story: As a business manager, I want to view sales analytics so that I can make data-driven decisions.

Acceptance Criteria:

- Dashboard loads within 3 seconds
- Data refreshes every 5 minutes automatically
- Graphs render correctly across browsers
- Exported reports include all filtered data

How It Must Perform — Non-Functional Requirements

NFR-1: Performance Requirements

- Homepage loads in 2 seconds on a 4G connection
- Product listing pages load within 3 seconds
- Checkout completes in under 5 seconds
- Handles 1,000 concurrent users without slowing down
- Supports peak load of 10,000 transactions per day
- Database queries execute within 500ms
- API responses under 200ms for 95% of requests

NFR-2: Security Requirements

- All data encrypted in transit using TLS 1.3
- Passwords hashed with bcrypt (minimum cost factor 12)
- CSRF protection on all forms
- SQL injection prevented through parameterized queries
- Rate limiting: 100 API requests per minute per IP
- PCI-DSS compliant for payment processing
- Role-based access control (RBAC) implemented
- Security audit logs kept for 12 months
- Sensitive data masked in logs

NFR-3: Availability and Reliability

- 99.5% uptime (excluding planned maintenance)
- Maintenance scheduled for off-peak hours (2–4 AM IST)
- Automated database backups every 6 hours
- Disaster recovery with 4-hour Recovery Point Objective
- Health check endpoints for monitoring
- Graceful handling of payment gateway failures

NFR-4: Scalability Requirements

- Horizontal scaling supported — can handle 3x traffic without code changes
- Database read replicas for query distribution
- Redis caching layer for frequently accessed data
- CDN for all static assets

NFR-5: Usability Requirements

- Fully responsive — works on desktop, tablet, and mobile (minimum 360x640 resolution)
- Compatible with Chrome, Firefox, Safari, and Edge (last 2 versions)
- Follows WCAG 2.1 Level AA accessibility standards
- Friendly, clear error messages throughout
- Checkout completes in maximum 3 steps

NFR-6: Maintainability Requirements

- Code follows ESLint style guide for JavaScript
- Minimum 80% code coverage in unit tests
- Comprehensive logging with severity levels
- API documentation using Swagger/OpenAPI
- Version control with Git using feature branch workflow

NFR-7: Compliance Requirements

- Complies with RBI payment guidelines
- GST calculation follows Indian tax regulations
- Complies with Indian IT Act 2000 for data protection

- Mandatory legal pages displayed (Privacy Policy, Terms of Service, Refund Policy)
- User consent obtained for data collection (cookies, marketing)

How Customers Will Use It

Journey 1: A First-Time Buyer (Guest Checkout)

1. Finds us on Google and lands on the homepage
2. Browses categories or searches for a specific product
3. Reads reviews, checks delivery estimate for their pincode
4. Adds to cart and heads to checkout
5. Enters email and delivery address — no account required
6. Picks a payment method, completes payment
7. Gets an order confirmation email instantly
8. Tracks their order via the link in the email
9. Receives their product
10. Gets a review request email 7 days later

Key improvements we're addressing:

- Simplify product discovery with smart search
- Minimize checkout fields
- Provide clear delivery estimates
- Send proactive order updates

Journey 2: A Returning Customer

1. Logs in and sees personalized recommendations
2. Finds their product and adds it to cart
3. Selects their saved address and preferred payment method
4. Completes the purchase in under 2 minutes
5. Tracks the order from "My Orders"

Key improvements we're addressing:

- Enable one-click reorder from order history
- Implement smart recommendations
- Support quick checkout for logged-in users

Business Rules to Keep in Mind

1. Minimum order value is ₹99
2. Free shipping on orders above ₹499
3. Orders can be cancelled within 24 hours if not yet shipped
4. Refunds process within 5–7 business days after approval
5. Returns accepted within 7 days of delivery — unused, original packaging only
6. Exchanges are only offered if the replacement is in stock
7. Promo codes cannot stack with sale prices
8. Maximum 5 units of the same product per order (anti-hoarding)
9. GST rates: 5% essential goods, 12% general, 18% electronics, 28% luxury
10. Accounts lock after 5 failed logins; unlock after 30 minutes or via email
11. Reviews only from verified purchasers, within 90 days of delivery
12. Product images must be JPEG or PNG, max 2MB
13. Guest users get a tracking link per order; full history requires an account

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Are We Succeeding? Key Performance Indicators

Metric	Current	Target (6M)	Measurement
Conversion Rate	45%	70%	Google Analytics
Cart Abandonment	70%	50%	System analytics
Order Processing Time	4 hours	30 min	Order dashboard
Order Accuracy	92%	98%	Fulfillment reports
Customer NPS	38	60+	Post-delivery survey
Page Load Time	5 sec	2 sec	GTmetrix/Lighthouse
Payment Success Rate	88%	95%	Payment gateway
System Uptime	97%	99.5%	Uptime monitoring
Daily Order Capacity	3,000	10,000	Load testing

Table 2: Key Performance Indicators and targets

Timeline: How We Get There

Phase	Duration	Key Deliverables
Requirements Gathering	Weeks 1–2	BRD, Wireframes, Technical specs
Design Phase	Weeks 3–4	UI/UX mockups, Database design
Development Phase 1	Weeks 5–10	Core functionality (catalog, cart, checkout)
Development Phase 2	Weeks 11–16	Admin panel, order management
Payment Integration	Weeks 15–17	Gateway integration and testing
Testing (QA)	Weeks 18–20	Functional, performance, security testing
User Acceptance Testing	Weeks 21–22	UAT with business stakeholders
Deployment	Week 23	Production deployment, monitoring setup
Post-Launch Support	Weeks 24–26	Bug fixes, performance optimization

Table 3: Project timeline with major milestones

Critical Path Items: Payment gateway integration, inventory management module, and performance optimization need special attention — any delay here cascades across the whole project.

The Numbers: Cost vs Benefit

What We're Spending

Cost Category	Amount (₹)
Development Team (6 months)	32,00,000
QA and Testing	6,00,000
UI/UX Design	4,00,000
Infrastructure (AWS hosting)	3,00,000
Payment Gateway Setup	1,00,000
Licensing and Tools	2,00,000
Contingency (10%)	2,00,000
Total	**50,00,000**

Table 4: Project cost breakdown

What We Stand to Gain (Annually)

- **₹1.2 crore** in additional revenue from better conversion and retention (40% increase)
- **₹15 lakh** saved from reducing manual processing costs (50% reduction)
- **₹8 lakh** saved by cutting fulfillment errors
- **₹20 lakh** from higher repeat purchases due to better customer experience (25% increase in CLV)

Total Annual Benefit: ₹1.63 crore
ROI: 180% within 18 months
Payback Period: 11 months

Risks and How We'll Handle Them

Risk	Likelihood	Our Mitigation Plan
Payment gateway integration delays	Medium	Start early, keep multiple gateway options ready
Data migration issues	High	Run pilot migration first, build in buffer time
Performance problems under peak traffic	Medium	Load-test early, use auto-scaling
Scope creep from stakeholders	High	Enforce change control, push extras to Phase 2
Team availability issues	Medium	Cross-train members, maintain documentation
Third-party API changes	Low	Abstract integrations, monitor for updates
Security vulnerabilities	Medium	Conduct security audit, automated scans
Budget overrun	Low	Track weekly spend, freeze scope after design

Table 5: Risk register with mitigation plans

What Needs to Be in Place Before We Can Build

Internal Dependencies

- Product catalog data cleaned and ready for migration
- Customer database accessible for migration
- ERP system API documentation provided by IT team
- SSL certificate and domain configuration completed
- AWS account setup with appropriate permissions

External Dependencies

- Payment gateway sandbox environments available (Razorpay, PayU, Paytm)
- Shipping partner API access and documentation (Delhivery, BlueDart)
- Email service provider configured (SendGrid or AWS SES)
- SMS gateway setup (MSG91 or Twilio)
- CDN provider configuration (Cloudflare)

Approval and Sign-Off

This Business Requirements Document requires approval from the following stakeholders:

Name	Role	Signature/Approval
	CEO/Project Sponsor	
	VP Operations	
	IT Manager	
	Finance Controller	
	Business Analyst	

Table 6: Stakeholder approval matrix

Approval Date: _____

Note: Any changes to this document after approval must go through the formal change control process.

Appendices

Appendix A: Glossary of Key Terms

- **SKU (Stock Keeping Unit):** A unique code for each distinct product variant
- **NPS (Net Promoter Score):** Customer satisfaction metric ranging from -100 to +100

- **RPO (Recovery Point Objective):** Maximum acceptable data loss measured in time
- **CSRF (Cross-Site Request Forgery):** Type of security vulnerability
- **PCI-DSS:** Payment Card Industry Data Security Standard
- **GST (Goods and Services Tax):** Indian indirect tax system
- **RBI (Reserve Bank of India):** Central banking institution regulating payments
- **AWB (Airway Bill):** Tracking number for shipped packages
- **CAGR (Compound Annual Growth Rate):** Year-over-year growth rate
- **UAT (User Acceptance Testing):** Final testing phase with business stakeholders

Appendix B: Related Documents

- Technical Design Document (TDD)
- System Architecture Document
- API Documentation (Swagger)
- UI/UX Wireframes and Mockups
- Test Strategy Document
- Deployment Plan
- User Training Manual

Appendix C: Document Change Log

Version	Date	Author	Changes
1.0	March 15, 2026	Business Analyst	Initial document creation

Table 7: Document revision history

END OF DOCUMENT